

Maharashtra Public Service Commission Assessment

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Assistant Professor Mathematics Group A and Lecturer Mathematics Group B

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Question Number : 1 Question Id : 4410092107454 Question Type : MCQ Option Shuffling : No

Which of the following best describes the convergence of a sequence of real numbers?

Options :

1. The terms approach a fixed real value as the sequence progresses
2. The terms become arbitrarily large as the sequence progresses
3. The terms remain within a certain fixed interval but do not settle
4. The terms repeat in a regular pattern without settling

Question Number : 2 Question Id : 4410092107462 Question Type : MCQ Option Shuffling : No

What is the sequence of partial sums for a given series of real numbers?

Options :

1. A sequence where each term is the n th term of the series itself
2. A sequence where each term is the sum of the first n terms of the series
3. A sequence where each term is the average of the first n terms of the series
4. A sequence where each term is the product of the first n terms of the series

Question Number : 3 Question Id : 4410092107465 Question Type : MCQ Option Shuffling : No

Which of the following best describes the real number system in terms of its structure?

Options :

1. A field without order
2. An ordered field without completeness
3. A complete ordered field
4. An incomplete ordered field

Question Number : 4 Question Id : 4410092107444 Question Type : MCQ Option Shuffling : No

If a function f is monotonic on an interval, which of the following statements is always true for any two points x and y in that interval where $x < y$?

Options :

1. Either $f(x) \leq f(y)$ for all x, y or $f(x) \geq f(y)$ for all x, y
2. $f(x) < f(y)$ for all x, y
3. $f(x) = f(y)$ for all x, y
4. $f(x) > f(y)$ for all x, y

Question Number : 5 Question Id : 4410092107446 Question Type : MCQ Option Shuffling : No

Suppose f and g are functions continuous on $[a, b]$ and differentiable on (a, b) , with $g'(x)$ not equal to zero on (a, b) . According to Cauchy's Mean Value Theorem, which of the following expressions is guaranteed to be equal for some c in (a, b) ?

Options :

1. $f'(c)g'(c) = (f(b) - f(a))(g(b) - g(a))$
2. $f'(c) + g'(c) = f(b) - f(a) + g(b) - g(a)$
3. $f'(c) - g'(c) = (f(b) - f(a)) - (g(b) - g(a))$
4. $\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$

Question Number : 6 Question Id : 4410092107463 Question Type : MCQ Option Shuffling : No

Which of the following statements correctly describes the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ for real $p > 0$?

Options :

1. The series diverges for all $p > 0$.
2. The series converges for all $p > 0$.
3. The series converges only if $p \geq 2$.
4. The series converges if $p > 1$ and diverges if $0 < p \leq 1$.

Question Number : 7 Question Id : 4410092107469 Question Type : MCQ Option Shuffling : No

Which statement best describes the level curves of a function $f(x, y)$ of two variables?

Options :

1. They are curves in the xy -plane where the function takes the same value.
2. They are curves in three-dimensional space where the function is plotted.
3. They are lines where the gradient of the function is zero.
4. They are points where both partial derivatives vanish.

Question Number : 8 Question Id : 4410092107479 Question Type : MCQ Option Shuffling : No

Which of the following statements best describes a metric space?

Options :

1. A set with a distance function that is only non-negative
2. A set with a distance function satisfying positivity, symmetry, and triangle inequality
3. A set with an order relation that is transitive and antisymmetric
4. A set with a binary operation that is commutative and associative

Question Number : 9 Question Id : 4410092107443 Question Type : MCQ Option Shuffling : No

Suppose f is a real-valued function defined on $[a, b]$ that is of bounded variation. Which of the following statements most accurately evaluates a possible property of f ?

Options :

1. f can be expressed as the difference of two increasing functions on $[a, b]$.
2. f is necessarily monotonic on $[a, b]$.
3. f must be continuous everywhere on $[a, b]$.
4. f must be differentiable at all points of $[a, b]$.

Question Number : 10 Question Id : 4410092107447 Question Type : MCQ Option Shuffling : No

Suppose a function f is continuous on $[1, 4]$ and differentiable on $(1, 4)$, with $f(1) = 3$ and $f(4) = 15$. Which of the following statements best illustrates a valid application of Lagrange's Mean Value Theorem to f on $[1, 4]$?

Options :

1. There exists a c in $(1, 4)$ such that $f'(c)$ equals 4.
2. The theorem does not apply to this function.
3. There exists a c in $(1, 4)$ such that $f'(c)$ equals 0.
4. For all c in $(1, 4)$, $f'(c)$ is greater than 4.

Question Number : 11 Question Id : 4410092107453 Question Type : MCQ Option Shuffling : No

Which of the following statements is the standard formulation of the Bolzano-Weierstrass Theorem for sequences in $\mathbb{R}$?

Options :

1. Every bounded sequence is convergent.
2. Every bounded sequence has at least one convergent subsequence.
3. Every sequence has a convergent subsequence.
4. Every Cauchy sequence is bounded.

Question Number : 12 Question Id : 4410092107456 Question Type : MCQ Option Shuffling : No

Suppose you have a sequence of closed and bounded intervals in the real numbers, where each interval is nested within the previous one, and the lengths of the intervals approach zero. Based on the Heine-Borel Theorem, what can you conclude about the intersection of all these intervals?

Options :

1. The intersection contains an open interval
2. The intersection contains exactly one point
3. The intersection contains more than one point
4. The intersection is empty

Question Number : 13 Question Id : 4410092107461 Question Type : MCQ Option Shuffling : No

A sequence is defined by $s_n = (-1)^n(1 + 1/n)$. Which of the following best describes the behavior of the series $\sum_n s_n$ as n approaches infinity?

Options :

1. The series is convergent
2. The series is conditionally convergent
3. The series is divergent and oscillatory
4. The series is absolutely convergent

Question Number : 14 Question Id : 4410092107474 Question Type : MCQ Option Shuffling : No

Suppose $F(x,y,z) = x^2 + y^2 + z^2 - 1$ defines a surface in $\mathbb{R}^3$. At the point $(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$, which of the following statements best describes the existence of a differentiable implicit function $z = f(x, y)$ near this point according to the implicit function theorem?

Options :

1. A differentiable implicit function $z = f(x, y)$ exists only if F is linear in z , which is not the case here.
2. A differentiable implicit function $z = f(x, y)$ does not exist because the partial derivative of F with respect to z is zero at the point.
3. No differentiable implicit function $z = f(x, y)$ exists near the point because F is quadratic in all variables.
4. A differentiable implicit function $z = f(x, y)$ exists near the point because the partial derivative of F with respect to z is nonzero there.

Question Number : 15 Question Id : 4410092107480 Question Type : MCQ Option Shuffling : No

In a normed vector space, which of the following statements best distinguishes compactness from sequential compactness in the context of infinite-dimensional spaces?

Options :

1. Total boundedness and completeness always imply compactness in normed spaces
2. Total boundedness and completeness always not imply compactness in normed spaces
3. Sequential compactness does not imply compactness in infinite-dimensional normed spaces
4. Heine-Borel theorem holds in all normed spaces

Question Number : 16 Question Id : 4410092107509 Question Type : MCQ Option Shuffling : No

Which of the following statements is true regarding the analysis of singularities using Laurent series in complex analysis?

Options :

1. The Laurent series always converges outside the neighborhood of the singularity.
2. A zero principal part in the Laurent series implies the singularity is essential.
3. A function's behavior near an isolated singularity can be determined by the principal part of its Laurent series expansion.
4. If the Laurent series has only non-negative powers, the singularity is essential.

Question Number : 17 Question Id : 4410092107513 Question Type : MCQ Option Shuffling : No

Determine the radius of convergence for a power series $\sum_{n=0}^{\infty} \frac{1}{n^n} z^n$ in complex analysis.

Options :

1. ∞
2. 0
3. 1
4. can not be determined

Question Number : 18 Question Id : 4410092107497 Question Type : MCQ Option Shuffling : No

Which of the following best describes the winding number of a closed curve about a point in the complex plane?

Options :

1. It measures the distance from the curve to the point.
2. It counts how many times the curve encircles the point, considering orientation.
3. It gives the area inside the curve with respect to the point.
4. It measures the length of the curve around the point.

Question Number : 19 Question Id : 4410092107501 Question Type : MCQ Option Shuffling : No

Which of the following conditions must be satisfied for a function $f(z) = u(x, y) + iv(x, y)$ to be analytic at a point in the complex plane, according to the Cauchy-Riemann equations?

Options :

1. The mixed partial derivatives u_{xy} and v_{yx} must be equal at the point
2. The partial derivatives u_x, u_y, v_x, v_y exist and $u_y = v_x$
3. The partial derivatives u_x and v_x exist but their equality is not required
4. The partial derivatives u_x, u_y, v_x and v_y exist with $u_x = v_y$, and $u_y = -v_x$

Question Number : 20 Question Id : 4410092107505 Question Type : MCQ Option Shuffling : No

Which condition must be satisfied for a bilinear transformation $f(z)$ to be conformal mapping in complex analysis?

Options :

1. The first order derivative is zero
2. The second order derivative is zero
3. The second order derivative is non-zero
4. The first order derivative is non-zero

Question Number : 21 Question Id : 4410092107486 Question Type : MCQ Option Shuffling : No

Suppose a function $f(z)$ has a simple pole at $z = 2$ and a removable singularity at $z = -1$ within a positively oriented simple closed contour C . Which statement best describes the value of the integral of $f(z)$ around C ?

Options :

1. The integral depends only on the residue at $z = 2$
2. The integral depends only on the residue at $z = -1$
3. The integral is always zero since there is a removable singularity
4. The integral is always one since there is a removable singularity

Question Number : 22 Question Id : 4410092107494 Question Type : MCQ Option Shuffling : No

Let $f(z)$ be an analytic function on and inside the closed disk $|z| \leq 2$. If $|f(z)|$ attains its maximum at some point z_0 with $|z_0| < 2$, which conclusion is valid about $f(z)$?

Options :

1. $f(z)$ must be zero at the boundary
2. $f(z)$ must attain its minimum at the point outside the region
3. $f(z)$ must have a maximum only on the boundary
4. $f(z)$ must be constant throughout the disk

Question Number : 23 Question Id : 4410092107498 Question Type : MCQ Option Shuffling : No

Suppose a function f is defined on the complex plane by $f(z) = az + b$, where a and b are complex numbers, and a is not zero. Which of the following statements best describes the geometric transformation performed by f on any complex number z ?

Options :

1. It performs a rotation and scaling followed by a translation.
2. It reflects points across the real axis.
3. It only rotates points around the origin.
4. It only scales points without changing their direction.

Question Number : 24 Question Id : 4410092107508 Question Type : MCQ Option Shuffling : No

Suppose a bilinear transformation maps the points 1 , -1 , and i in the z -plane to the points 0 , ∞ , and 1 in the w -plane, respectively. Which of the following is the normal form of this bilinear transformation?

Options :

1. $w = -i \frac{z-1}{z+1}$
2. $w = \frac{z+1}{z-1}$
3. $w = \frac{z+i}{z-i}$
4. $w = \frac{z-i}{z+i}$

Question Number : 25 Question Id : 4410092107512 Question Type : MCQ Option Shuffling : No

Suppose $f(z)$ is analytic in the annulus $1 < |z| < 3$. Which statement best describes how the Laurent series for $f(z)$ about $z = 0$ can be used to evaluate the contour integral of $f(z)$ around the circle $|z| = 2$?

Options :

1. The integral equals $2\pi i$ times the sum of all negative-power coefficients in the Laurent series
2. The integral is zero since $f(z)$ is analytic in the annulus
3. The integral equals $2\pi i$ times the coefficient of z in the Laurent series
4. The integral equals $2\pi i$ times the coefficient of $1/z$ in the Laurent series

Question Number : 26 Question Id : 4410092107556 Question Type : MCQ Option Shuffling : No

Which statement correctly describes Euler's phi-function for a prime number p ?

Options :

1. It equals p
2. It equals zero
3. It equals one
4. It equals p minus one

Question Number : 27 Question Id : 4410092107566 Question Type : MCQ Option Shuffling : No

What is the number of elements in a finite field called?

Options :

1. The characteristic of the field
2. The modulus of the field
3. The degree of the field
4. The order of the field

Question Number : 28 Question Id : 4410092107558 Question Type : MCQ Option Shuffling : No

Which of the following statements accurately describes a group in the context of algebraic structures?

Options :

1. A set with a binary operation and an identity element, but not necessarily associativity
2. A set with a binary operation that is associative, has an identity element, and each element has an inverse
3. A set with a commutative operation and an identity element
4. A set with a binary operation where only some elements have inverses

Question Number : 29 Question Id : 4410092107561 Question Type : MCQ Option Shuffling : No

Which statement best describes the conclusion of Sylow's First Theorem for a finite group G of order 56?

Options :

1. There exists a subgroup of order 7 in G
2. Every subgroup of order 7 is normal in G
3. There exists a unique subgroup of order 8 in G
4. There exists a normal subgroup of order 14 in G

Question Number : 30 Question Id : 4410092107572 Question Type : MCQ Option Shuffling : No

Which of the following statements correctly characterizes separable extensions in the context of field theory?

Options :

1. A finite extension is separable if every polynomial splits into linear factors in the extension.
2. Separable extensions only exist in characteristic zero fields.
3. A finite extension is separable if every element has a separable minimal polynomial over the base field.
4. A finite extension is separable if all irreducible polynomials are reducible in the extension.

Question Number : 31 Question Id : 4410092107546 Question Type : MCQ Option Shuffling : No

A research lab needs to assign 4 distinct research projects to 5 qualified scientists. Each project must be assigned to exactly one scientist, and each scientist can handle at most one project. In how many different ways can the projects be assigned?

Options :

1. 120
2. 24
3. 20
4. 625

Question Number : 32 Question Id : 4410092107557 Question Type : MCQ Option Shuffling : No

Which of the following best describes the number of distinct primitive roots modulo 26?

Options :

1. 4
2. 12
3. 7
4. 15

Question Number : 33 Question Id : 4410092107562 Question Type : MCQ Option Shuffling : No

Let R be a commutative ring with unity, and let I and J be ideals of R such that $I + J = R$. Which of the following statements is necessarily true?

Options :

1. $I \cap J = I + J$
2. $I \cap J = \{0\}$
3. $I \cap J = IJ$
4. $IJ = R$

Question Number : 34 Question Id : 4410092107564 Question Type : MCQ Option Shuffling : No

Let R be a commutative ring with unity and let I be an ideal of R . If every element r in R satisfies either $r \in I$ or r is a unit modulo I , what can be concluded about I ?

Options :

1. I is a principal ideal
2. I is a prime ideal
3. I is a maximal ideal
4. I is a minimal ideal

Question Number : 35 Question Id : 4410092107569 Question Type : MCQ Option Shuffling : No

Suppose K is a field and E is a finite extension of K such that every irreducible polynomial in $K[x]$ with a root in E splits completely in E . Which statement best describes E over K ?

Options :

1. E is a simple extension of K
2. E is a purely inseparable extension of K
3. E is a normal extension of K
4. E is a Galois extension of K

Question Number : 36 Question Id : 4410092107593 Question Type : MCQ Option Shuffling : No

Which of the following best describes the first countability axiom in topology?

Options :

1. The space has a countable base.
2. Every subset is countable.
3. Every point has a countable local base.
4. The space is compact.

Question Number : 37 Question Id : 4410092107578 Question Type : MCQ Option Shuffling : No

If $X = \{a, b, c\}$ and $B = \{\{a\}, \{a, b\}\}$, then which of the following topology on X is generated by B ?

Options :

1. $\{\emptyset, \{a\}, \{b\}, \{c\}, X\}$
2. The discrete topology on X
3. $\{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$
4. $\{\emptyset, \{a\}, \{a, b\}, X\}$

Question Number : 38 Question Id : 4410092107581 Question Type : MCQ Option Shuffling : No

Let X and Y be topological spaces. Which of the following statements correctly describes the open sets in the product topology on $X \times Y$?

Options :

1. Open sets are arbitrary intersections of open sets from X and Y
2. Open sets are arbitrary unions of products of open sets from X and Y
3. Open sets are only products of open sets from X and Y
4. Open sets must be open in both X and Y simultaneously

Question Number : 39 Question Id : 4410092107577 Question Type : MCQ Option Shuffling : No

Let X be an infinite set with the cofinite topology and Y is a set with the discrete topology. Let $f : X \rightarrow Y$ be a function. Under what condition is f continuous?

Options :

1. f must be injective
2. f must be surjective
3. f must be constant on X
4. f must be bijective

Question Number : 40 Question Id : 4410092107582 Question Type : MCQ Option Shuffling : No

Suppose $X = Y = \mathbb{R}$, the set of all real numbers and X is given the discrete topology while Y is given the indiscrete topology. Consider the product space $X \times Y$ with the product topology. Which statement is true in $X \times Y$?

Options :

1. The closed unit disk in $X \times Y$ is open in the product topology.
2. Every subset of $X \times Y$ is open in the product topology.
3. The vertical line $\{ (2, y) : y \text{ is a real number} \}$ is open in the product topology
4. The horizontal line $\{ (x, 1) : x \text{ is a real number} \}$ is open in the product topology

Question Number : 41 Question Id : 4410092107585 Question Type : MCQ Option Shuffling : No

Suppose Y is a subspace of a topological space X , and A is a subset of Y . Under what condition is the subspace topology on A inherited from Y the same as the subspace topology on A inherited directly from X ?

Options :

1. The subspace topology on A inherited from Y is the same only if Y is closed in X .
2. The subspace topology on A inherited from Y is the same only if Y is open in X .
3. The subspace topology on A inherited from Y coincides only if every open set in X is open in Y .
4. The subspace topology on A inherited from Y is always the same as that inherited directly from X .

Question Number : 42 Question Id : 4410092107590 Question Type : MCQ Option Shuffling : No

In the co-countable topology on the set of all real numbers $\mathbb{R}$, what is the set of limit points of $[0,1]$?

Options :

1. The set $[0, 1]$ itself
2. The set $\{0,1\}$
3. The set of all real numbers $\mathbb{R}$
4. The set $(0, 1)$

Question Number : 43 Question Id : 4410092107599 Question Type : MCQ Option Shuffling : No

Which of the following is a defining property of a sigma field in probability theory?

Options :

1. It contains at least two nonempty set
2. It is closed under countable unions of its sets
3. It is closed under finite intersections only
4. It is closed under symmetric difference

Question Number : 44 Question Id : 4410092107619 Question Type : MCQ Option Shuffling : No

Which of the following best describes the Kolmogorov strong law of large numbers?

Options :

1. It states that the sample average converges in probability to the expected value as the sample size increases.
2. It states that the sample variance converges almost surely to the expected value as the sample size increases.
3. It states that the sample sum remains constant as the sample size increases.
4. It states that the sample average converges almost surely to the expected value as the sample size increases.

Question Number : 45 Question Id : 4410092107597 Question Type : MCQ Option Shuffling : No

If a probability space is defined on the set $S = \{1, 2, 3, 4, 5, 6\}$ with probability assigned as $P(\{1\}) = 0.1$, $P(\{2\}) = 0.1$, $P(\{3\}) = 0.1$, $P(\{4\}) = 0.2$, $P(\{5\}) = 0.2$, what is the probability $P(\{2,4,6\})$?

Options :

1. 0.3
2. 0.2
3. 0.6
4. 0.4

Question Number : 46 Question Id : 4410092107605 Question Type : MCQ Option Shuffling : No

Suppose a test for a disease is 95% accurate. If 2% of the population has the disease, what is the probability that a randomly selected person who tests positive actually has the disease?

Options :

1. 0.95
2. 0.28
3. 0.02
4. 0.67

Question Number : 47 Question Id : 4410092107600 Question Type : MCQ Option Shuffling : No

Which of the following collections is NOT a sigma-field of subsets of the real line?

Options :

1. The collection of all subsets of the real line
2. The collection of all Borel sets
3. The collection of all open intervals in the real line
4. The collection of all intervals, their complements and countable union of intervals and their complements

Question Number : 48 Question Id : 4410092107610 Question Type : MCQ Option Shuffling : No

A machine produces items with a probability of 0.1 for a defect. If 12 items are produced, what is the probability that exactly 2 are defective?

Options :

1. 0.1086
2. 0.3874
3. 0.0523
4. 0.2301

Question Number : 49 Question Id : 4410092107612 Question Type : MCQ Option Shuffling : No

A pharmaceutical company is comparing the heights of two populations: Population A follows a normal distribution with mean 170 cm and standard deviation 6 cm, and Population B follows a normal distribution with mean 175 cm and standard deviation 8 cm. If a randomly selected individual from each population is taller than 180 cm, which population is more likely to produce such an individual?

Options :

1. Population B is more likely
2. There is insufficient information to decide
3. Population A is more likely
4. Both populations are equally likely

Question Number : 50 Question Id : 4410092107614 Question Type : MCQ Option Shuffling : No

Suppose X and Y are independent random variables, where X follows a discrete probability distribution with finite mean and variance, and Y follows an absolutely continuous probability distribution with finite mean and variance. If a new random variable Z is defined as $Z = 2X - 3Y + 4$, Which of the following statements about the variance of Z is true?

Options :

1. $\text{Var}(Z) = 4 \text{Var}(X) + 9 \text{Var}(Y)$
2. $\text{Var}(Z) = 2 \text{Var}(X) - 3 \text{Var}(Y)$
3. $\text{Var}(Z) = 2 \text{Var}(X) - 3 \text{Var}(Y) + 4$
4. $\text{Var}(Z) = 4 \text{Var}(X) + 9 \text{Var}(Y) + 16$

Question Number : 51 Question Id : 4410092107706 Question Type : MCQ Option Shuffling : No

Which of the following methods is commonly used to solve a first-order linear ordinary differential equation?

Options :

1. Newton–Raphson method
2. Variation of parameters
3. Integrating factor method
4. Lagrange multipliers method

Question Number : 52 Question Id : 4410092107701 Question Type : MCQ Option Shuffling : No

Which of the following statements best describes a singular solution of a higher degree ordinary differential equation (ODE)?

Options :

1. It is a solution that cannot be obtained from the general solution by assigning any value to the arbitrary constants.
2. It is a solution corresponding to a specific initial value of the dependent variable only.
3. It is the only solution that passes through every point in the plane.
4. It is a solution that is valid for all possible functions of the independent variable.

Question Number : 53 Question Id : 4410092107704 Question Type : MCQ Option Shuffling : No

Which of the following equations best represents a second-order linear ordinary differential equation with variable coefficients?

Options :

1. $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$
2. $y \frac{dy}{dx} + x^2 = 0$
3. $\frac{d^2y}{dx^2} + 4 \frac{dy}{dx} + 3y = 0$
4. $\frac{d^2y}{dx^2} + \sin(y) = 0$

Question Number : 54 Question Id : 4410092107700 Question Type : MCQ Option Shuffling : No

A first-order higher-degree differential equation exhibits a feature that distinguishes it from a first-order first-degree differential equation. Which of the following statements best describes this feature?

Options :

1. It can always be reduced to a linear differential equation by a suitable substitution.
2. It cannot be solved unless it is homogeneous.
3. It may possess multiple solution branches corresponding to different values of the derivative at a point.
4. Its general solution always contains exactly one arbitrary constant and no singular solution.

Question Number : 55 Question Id : 4410092107703 Question Type : MCQ Option Shuffling : No

Given the non-homogeneous linear differential equation $y'' - 4y' + 4y = e^{2x}$, which of the following is the most suitable method for obtaining a particular solution?

Options :

1. Variation of parameters for arbitrary right-hand sides
2. Undetermined coefficients using a polynomial times e^{2x}
3. Laplace transform to solve the equation
4. Direct substitution of an arbitrary function

Question Number : 56 Question Id : 4410092107709 Question Type : MCQ Option Shuffling : No

A second-order linear nonhomogeneous differential equation $y'' + p(x)y' + q(x)y = g(x)$ is given. Which step is essential when using the method of variation of parameters to construct a particular solution?

Options :

1. Substituting arbitrary differentiable functions for the solution
2. Using the homogeneous solution with constant coefficients only
3. Integrating the nonhomogeneous term directly without involving homogeneous solutions
4. Expressing the particular solution as a linear combination of the homogeneous solutions with variable coefficients

Question Number : 57 Question Id : 4410092107716 Question Type : MCQ Option Shuffling : No

A scientist models the cooling of a metal rod using the differential equation $\frac{dy}{dt} + 3y = 15$, where y is the temperature at time t . She wants to use the integrating factor method to solve it. Which of the following should be done first after identifying it as a linear differential equation and finding its integrating factor?

Options :

1. Divide both sides by y
2. Multiply both sides by e^{3t}
3. Take the derivative of both sides
4. Add 3 to both sides

Question Number : 58 Question Id : 4410092107732 Question Type : MCQ Option Shuffling : No

Which of the following is a valid first step in applying the method of separation of variables to solve Laplace's equation in two variables?

Options :

1. Guess a solution that satisfies the Laplace's equation
2. Assume that the solution can be written as a product of two functions, each depending on a single variable
3. Integrate both sides with respect to one variable
4. State all boundary conditions before manipulating the equation

Question Number : 59 Question Id : 4410092107718 Question Type : MCQ Option Shuffling : No

Which of the following pairs correctly matches a type of second-order partial differential equation (PDE) with its characteristic form?

Options :

1. Elliptic PDE – two sets of real characteristic curves
2. Elliptic PDE – no real characteristic curves
3. Hyperbolic PDE – no real characteristic curves
4. Parabolic PDE – two sets of real characteristic curves

Question Number : 60 Question Id : 4410092107733 Question Type : MCQ Option Shuffling : No

Which form of solution is typically assumed when applying the method of separation of variables to solve the one-dimensional heat equation?

Options :

1. A function of the sum of the independent variables
2. A sum of two functions, each depending on a single variable
3. A single exponential function involving both variables
4. A product of two functions, each depending on a single variable

Question Number : 61 Question Id : 4410092107719 Question Type : MCQ Option Shuffling : No

A researcher encounters a fourth-order partial differential equation arising from a physical model. The principal part has the form $u_{xxxx} - 5u_{xxyy} + 4u_{yyyy} = 0$. Which classification best describes this PDE based on its principal symbol?

Options :

1. Elliptic
2. Strictly hyperbolic
3. Parabolic
4. Mixed type

Question Number : 62 Question Id : 4410092107725 Question Type : MCQ Option Shuffling : No

A first-order partial differential equation is given by $F(x, y, z, p, q) = 0$, where $p = \partial z / \partial x$ and $q = \partial z / \partial y$. If Charpit's method is applied to solve this PDE, which step correctly initiates the construction of the complete integral?

Options :

1. Form Charpit's auxiliary equations using the partial derivatives of F with respect to its variables.
2. Integrate F with respect to x while keeping y constant
3. Guess a particular solution based on the form of F
4. Substitute initial conditions directly into F to obtain a specific solution

Question Number : 63 Question Id : 4410092107734 Question Type : MCQ Option Shuffling : No

A metal rod is insulated along its lateral surface, and its temperature distribution is governed by the one-dimensional heat equation. Using the method of separation of variables, which of the following boundary conditions leads to an eigenfunction expansion containing only positive (nonzero) eigenvalues and sinusoidal eigenfunctions?

Options :

1. Both ends insulated (Neumann boundary conditions)
2. One end maintained at zero temperature and the other end insulated (mixed boundary conditions)
3. Both ends maintained at zero temperature (Dirichlet boundary conditions)
4. Both ends maintained at the same fixed nonzero temperature

Question Number : 64 Question Id : 4410092107740 Question Type : MCQ Option Shuffling : No

Which of the following is a necessary condition for a function $y(x)$ to provide an extremum to a functional in variational calculus?

Options :

1. The function must have a finite definite integral.
2. The function must satisfy the Euler-Lagrange equation.
3. The function must be differentiable at endpoints.
4. The function must be continuous everywhere.

Question Number : 65 Question Id : 4410092107738 Question Type : MCQ Option Shuffling : No

Which of the following best describes the role of the Euler-Lagrange equation in the calculus of variations?

Options :

1. It is used to directly solve partial differential equations.
2. It provides a necessary condition for a functional to have an extremum.
3. It always finds the maximum of a function regardless of the functional.
4. It guarantees the existence of a solution to a functional.

Question Number : 66 Question Id : 4410092107744 Question Type : MCQ Option Shuffling : No

In the calculus of variations for several variables, what does the Euler-Lagrange equation determine when applied to a functional defined over a multidimensional domain?

Options :

1. It evaluates the definite value of a multiple integral over a region
2. It guarantees the existence of an extremal solution for any functional
3. It describes the boundaries of the domain for the functional
4. It provides necessary conditions for a function to be an extremum of the functional

Question Number : 67 Question Id : 4410092107741 Question Type : MCQ Option Shuffling : No

A researcher is analyzing a functional $J[y] = \int_a^b F(x, y, y') dx$ and proposes that a function $y(x)$ is an extremal for J if it satisfies a necessary condition involving partial derivatives of F . Which approach best allows the researcher to verify that $y(x)$ satisfies the necessary condition for an extremum of J , considering admissible variations that vanish at the boundaries?

Options :

1. Verify that $y(x)$ satisfies the Euler-Lagrange equation and that variations vanish at the boundaries
2. Ensure the second derivative of F with respect to y' is continuous
3. Evaluate J at $y(x)$ and compare with neighboring functions
4. Check only the first derivative of $y(x)$ for zero at all points

Question Number : 68 Question Id : 4410092107745 Question Type : MCQ Option Shuffling : No

A researcher is optimizing a surface defined by a function of two variables using the calculus of variations. Which principle must be applied to find the necessary conditions for an extremum in the case of several variables?

Options :

1. The Euler–Ostrogradsky equation must be used for functionals involving only first-order derivatives
2. The method of Lagrange multipliers must be used without modification
3. The classical Euler-Lagrange equation must be used for each dependent variable
4. Green's theorem must be applied to reduce the surface integral

Question Number : 69 Question Id : 4410092107749 Question Type : MCQ Option Shuffling : No

A researcher is analyzing a variational problem where the functional $J[y] = \int (y'^2 + y^2) dx$ is to be minimized, subject to the subsidiary condition $\int y dx = c$ where c is a given constant. Which technique should be applied to solve this constrained variational problem?

Options :

1. Apply the Euler-Lagrange equation directly to the given functional
2. Introduce a Lagrange multiplier and form the augmented functional
3. Use only the boundary conditions to account for the constraint
4. Substitute the constraint directly into the functional before variation

Question Number : 70 Question Id : 4410092107765 Question Type : MCQ Option Shuffling : No

Which of the following best describes the motion of a rigid body in two dimensions?

Options :

1. It only involves motion along straight lines without any rotation.
2. It only involves rotation about a fixed axis in the plane.
3. It can be described as a combination of translation and rotation in a plane.
4. It is characterized by expansion or contraction within the plane.

Question Number : 71 Question Id : 4410092107758 Question Type : MCQ Option Shuffling : No

Which statement best captures the essence of Hamilton's canonical equations in analytical mechanics?

Options :

1. They provide second-order equations for coordinates using the Lagrangian function.
2. They relate total energy conservation to the kinetic energy only.
3. They express the time evolution of generalized coordinates and momenta using the Hamiltonian function.
4. They provide equations for only the generalized coordinates using the potential energy.

Question Number : 72 Question Id : 4410092107761 Question Type : MCQ Option Shuffling : No

Which statement accurately describes Hamilton's principle in analytical mechanics within the context of classical mechanics?

Options :

1. It asserts that the actual path taken by a system makes the action integral stationary over the considered time interval.
2. It requires that the system's velocity remains constant throughout its motion.
3. It implies that the potential energy of the system stays unchanged during its evolution.
4. It is satisfied only if the system's total energy is conserved at every instant.

Question Number : 73 Question Id : 4410092107757 Question Type : MCQ Option Shuffling : No

A bead of mass m moves without friction on a circular wire of radius R that rotates with constant angular velocity ω about a vertical diameter. Using Lagrange's equations, which statement best describes the equilibrium position(s) of the bead?

Options :

1. The equilibrium positions occur where the gravitational and centrifugal effects balance along the circular wire.
2. The equilibrium position is at the top of the circle regardless of the angular velocity.
3. The equilibrium position always corresponds to the minimum of the gravitational potential energy alone.
4. The bead remains only at the lowest point of the circle for any angular velocity.

Question Number : 74 Question Id : 4410092107762 Question Type : MCQ Option Shuffling : No

A particle is constrained to move along a frictionless surface under a conservative force field. Using Hamilton's principle, which of the following statements correctly evaluates how boundary conditions on the action functional impact the resulting equations of motion?

Options :

1. The action is stationary for paths with fixed initial and final positions, leading to the Euler-Lagrange equations.
2. The action must be maximized for arbitrary variations in both positions and velocities.
3. The action is stationary only if both positions and momenta are specified at the boundaries.
4. The action is stationary for any path, regardless of endpoint constraints.

Question Number : 75 Question Id : 4410092107768 Question Type : MCQ Option Shuffling : No

A symmetric top is spinning about its axis of symmetry with one point fixed. According to Euler's dynamical equations, which condition leads to steady precession without nutation?

Options :

1. The symmetry axis changes its inclination continuously with time.
2. All three principal moments of inertia are equal.
3. The angular velocity remains perpendicular to the symmetry axis.
4. The gravitational torque is exactly balanced by the gyroscopic effect, so that the inclination angle remains constant.

Question Number : 76 Question Id : 4410092107826 Question Type : MCQ Option Shuffling : No

Which of the following is always true about the eigenvalues of a square matrix?

Options :

1. They are always zero for any matrix.
2. They must always be integers for any matrix.
3. They may be real or complex depending on the matrix.
4. They are always positive for any matrix.

Question Number : 77 Question Id : 4410092107834 Question Type : MCQ Option Shuffling : No

Which statement best describes the Jordan canonical form of a square matrix?

Options :

1. It is a diagonal matrix with eigenvalues as entries only.
2. It is an upper-triangular matrix with zeros below the main diagonal.
3. It is a block diagonal matrix with each block corresponding to one eigenvalue.
4. It is a row-reduced echelon form used for solving systems.

Question Number : 78 Question Id : 4410092107836 Question Type : MCQ Option Shuffling : No

Which property must every inner product on a vector space satisfy?

Options :

1. It must be anti-symmetric for all vectors.
2. It must be symmetric for all vectors.
3. It must be linear in its second argument.
4. It must be linear in its first argument.

Question Number : 79 Question Id : 4410092107809 Question Type : MCQ Option Shuffling : No

If V and W are finite-dimensional vector spaces over the same field and T is a linear transformation from V to W , which statement correctly relates the dimension of the kernel of T , the dimension of the image of T , and the dimension of V ?

Options :

1. The difference between the dimensions of V and W equals the dimension of the kernel of T
2. The dimension of the image of T is always less than the dimension of the kernel of T
3. The sum of the dimensions of the kernel and image of T equals the dimension of W
4. The sum of the dimensions of the kernel and image of T equals the dimension of V

Question Number : 80 Question Id : 4410092107819 Question Type : MCQ Option Shuffling : No

Given a 3×3 matrix with two identical rows, what is the value of its determinant?

Options :

1. The determinant depends on the diagonal elements.
2. The determinant is zero.
3. The determinant depends on the specific entries.
4. The determinant is negative.

Question Number : 81 Question Id : 4410092107824 Question Type : MCQ Option Shuffling : No

Which of the following statements best describes the annihilating polynomial of a square matrix A ?

Options :

1. It is the characteristic polynomial of A with all roots being eigenvalues of A .
2. It must always have only real coefficients regardless of the entries of A .
3. Any polynomial $p(x)$ (not necessarily minimal) such that $p(A)=0$.
4. It is any polynomial that, when evaluated at A , yields the identity matrix.

Question Number : 82 Question Id : 4410092107806 Question Type : MCQ Option Shuffling : No

Which of the following statements best evaluates whether the set of all nonzero real numbers under usual addition forms a vector space over the field of real numbers?

Options :

1. The set fails to form a vector space because scalar multiplication is not defined.
2. The set fails to form a vector space because it is not closed under addition.
3. The set forms a vector space under the given operation.
4. The set fails to form a vector space because additive inverses do not exist.

Question Number : 83 Question Id : 4410092107811 Question Type : MCQ Option Shuffling : No

Let T be a linear transformation from $\mathbb{R}^4$ to $\mathbb{R}^3$ represented by a matrix A . If the dimension of the image of T is 2, which of the following must be true about the rank of T and the possible dimensions of the kernel of T ?

Options :

1. The rank of T is 3 and the dimension of the kernel of T is 1
2. The rank of T is 2 and the dimension of the kernel is at most 1
3. The rank of T is 2 and the dimension of the kernel of T is exactly 2
4. The rank of T is 4 and the kernel has dimension 0

Question Number : 84 Question Id : 4410092107813 Question Type : MCQ Option Shuffling : No

Suppose V and W are finite-dimensional vector spaces over the same field, and $T: V \rightarrow W$ is a linear transformation. Which scenario will guarantee that the nullity of T increases if a new vector is added to the basis of V , but T remains unchanged on all original basis vectors?

Options :

1. If the new vector is mapped to the zero vector in W and is linearly independent from the kernel of T
2. If the new vector is mapped to a nonzero vector in W
3. If T is redefined to map the new vector arbitrarily
4. If the new vector is a linear combination of existing kernel vectors

Question Number : 85 Question Id : 4410092107825 Question Type : MCQ Option Shuffling : No

Let T be a linear operator on a finite-dimensional vector space V such that its characteristic polynomial splits over the field F and T is diagonalizable. Which statement about the minimal polynomial of T is correct?

Options :

1. The minimal polynomial of T has distinct linear factors over F .
2. The minimal polynomial of T has a single irreducible quadratic factor over F .
3. The minimal polynomial of T may have repeated linear factors over F .
4. The minimal polynomial of T is always a product of distinct irreducible polynomials of degree greater than one.

Question Number : 86 Question Id : 4410092107839 Question Type : MCQ Option Shuffling : No

Given a quadratic form $Q(x, y, z) = 4x^2 + 6xy + 9y^2 + 8xz + 12yz + 16z^2$, which statement correctly describes the process to reduce Q to its canonical form?

Options :

1. Perform a sequence of orthogonal transformations to diagonalize the symmetric matrix associated with Q
2. Apply row operations to make the matrix triangular
3. Use variable substitution to set all cross-product terms to non-zero
4. Transform the matrix into an upper triangular form using elementary operations

Question Number : 87 Question Id : 4410092107841 Question Type : MCQ Option Shuffling : No

A researcher needs to classify a quadratic form $Q(x) = x^T A x$ in $\mathbb{R}^3$, where the matrix A is symmetric and has eigenvalues 4, -2, and 0. Which classification correctly describes $Q(x)$?

Options :

1. Positive definite
2. Semi-definite
3. Indefinite
4. Negative definite

Question Number : 88 Question Id : 4410092107848 Question Type : MCQ Option Shuffling : No

How many variables does the kernel $K(x, t)$ depend on in both Fredholm and Volterra integral equations?

Options :

1. 2
2. 0
3. 1
4. 3

Question Number : 89 Question Id : 4410092107843 Question Type : MCQ Option Shuffling : No

Which statement correctly describes a linear integral equation of the first kind of Fredholm type?

Options :

1. $u(x) = f(x) + \lambda \int_a^x K(x, t) u(t) dt$
2. $f(x) = u(x) + \int_a^b K(x, t) dt$
3. $u(x) = f(x) + \int_a^x K(x, t) u(t) dt$
4. $f(x) = \int_a^b K(x, t) u(t) dt$

Question Number : 90 Question Id : 4410092107855 Question Type : MCQ Option Shuffling : No

Which of the following best characterizes a singular integral equation within the context of linear integral equations?

Options :

1. It has a continuous kernel for all values in the integration range.
2. It contains a kernel that becomes infinite or undefined at certain points within the integration range.
3. It only contains algebraic singularities outside the range of integration.
4. It involves integration limits where the upper and lower bounds are variable.

Question Number : 91 Question Id : 4410092107844 Question Type : MCQ Option Shuffling : No

Consider the linear integral equation of the second kind of Fredholm type: $f(x) = g(x) + \lambda \int_a^b K(x, t) f(t) dt$. Which approach is most suitable for evaluating the existence of non-trivial solutions to this equation when $K(x, t)$ is continuous on $[a, b] \times [a, b]$ and λ is a parameter?

Options :

1. Analyze the characteristic values (eigenvalues) of the integral operator defined by $K(x, t)$
2. Check only the continuity of $g(x)$ on $[a, b]$
3. Directly differentiate the kernel with respect to x
4. Convert the equation to a Volterra type by changing limits

Question Number : 92 Question Id : 4410092107856 Question Type : MCQ Option Shuffling : No

Suppose the temperature distribution in a semi-infinite rod is modeled by

$$f(x) = \int_0^x \frac{\phi(t)}{\sqrt{x-t}} dt,$$

Identify the type of integral equation represented by this model.

Options :

1. Singular integral equation
2. Volterra integral equation of 1st kind
3. Fredholm integral equation
4. Abel integral equation

Question Number : 93 Question Id : 4410092107857 Question Type : MCQ Option Shuffling : No

Which of the following methods is commonly used for finding numerical solutions to algebraic equations?

Options :

1. Divided Difference Method
2. Newton-Raphson method
3. Runge-Kutta method
4. Trapezoidal rule

Question Number : 94 Question Id : 4410092107862 Question Type : MCQ Option Shuffling : No

Which method among the following is known for having quadratic rate of convergence when solving algebraic equations numerically?

Options :

1. Secant method
2. Bisection method
3. Fixed-point iteration
4. Newton-Raphson method

Question Number : 95 Question Id : 4410092107869 Question Type : MCQ Option Shuffling : No

In Hermite interpolation, which of the following best describes the type of data used to construct the interpolating polynomial?

Options :

1. Both function values and derivative values at the nodes
2. Only derivative values at the nodes
3. Only second derivative values at the nodes
4. Only function values at the nodes

Question Number : 96 Question Id : 4410092107878 Question Type : MCQ Option Shuffling : No

Which step is essential in the modified Euler method for numerically solving ordinary differential equations?

Options :

1. Evaluating the slope only at the midpoint
2. Calculating the average slope using initial and predicted values
3. Accepting the first predicted value as the solution
4. Using only the initial slope for prediction

Question Number : 97 Question Id : 4410092107858 Question Type : MCQ Option Shuffling : No

A researcher is tasked with solving a system of nonlinear algebraic equations numerically for a complex engineering model. She considers using the Newton-Raphson method but is concerned about convergence and computational cost. Which strategy would most effectively improve both convergence reliability and computational efficiency in this context?

Options :

1. Using only the Bisection method for all iterations
2. Applying only the Newton-Raphson method with a random initial guess
3. Increasing the step size in Newton-Raphson method
4. Implementing a hybrid approach that starts with the Bisection method and switches to Newton-Raphson near the root

Question Number : 98 Question Id : 4410092107873 Question Type : MCQ Option Shuffling : No

A researcher is analyzing experimental data where function values are recorded at equally spaced points, but the functional form is unknown. To estimate the derivative at an interior point with minimal bias and optimal use of nearby data, which numerical differentiation formula is most appropriate?

Options :

1. Higher-order forward difference formula
2. Central difference formula using adjacent points
3. Backward difference formula using right points
4. Forward difference formula using left points

Question Number : 99 Question Id : 4410092107877 Question Type : MCQ Option Shuffling : No

A researcher is using Euler's method to numerically solve the initial value problem $dy/dx = x + y$, $y(0) = 1$ with a step size $h = 0.1$ over the interval $[0, 0.2]$. After completing two steps, the researcher notices the computed y -value differs from the exact analytical solution at $x = 0.2$. Which best explains why this discrepancy occurs?

Options :

1. Euler's method introduces local truncation error at each step, causing cumulative deviation from the exact solution.
2. The step size changes after each iteration, leading to inconsistent results.
3. The exact solution is needed at every step to ensure accuracy, which Euler's method does not use.
4. The method fails to use the chain rule properly during iteration, causing error to appear.

Question Number : 100 Question Id : 4410092107879 Question Type : MCQ Option Shuffling : No

A research engineer is tasked with approximating the value of y at $x = 0.2$ for the initial value problem $dy/dx = x + y$, $y(0) = 1$, using the modified Euler method with a step size of 0.1. After the first step, which of the following best describes the method's approach to updating the solution for y at $x = 0.2$?

Options :

1. The slope at the midpoint of the interval is used to update y
2. Only the slope at the start of the interval is used to update y
3. The average of the slopes at the beginning and end of the interval is used to update y
4. Only the slope at the end of the interval is used to update y